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“What do they think about my idea?”- Co-Designing an E-Participation Tool for Participatory Urban Planning in Germany

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Abstract

This paper reports on multiple co-design workshops for the development of an e-participation tool. Participants were citizens and city staff from Wuppertal, Germany, and experts on participation. We explored what information citizens and city staff need, how this information should be provided, and what features for input and communication should be implemented. Through a thematic analysis, we found that information should be short, visual, and accessible. This visual information includes a visualization of the area relevant to participation. Participants preferred an abstract 3D model over a 2D map and a photorealistic 3D model. Additional visualizations should complement this abstract 3D model to support better recognition of the area, e.g. through photos and videos, or through an extra layer that visualizes the terrain. Three preferred ways of interaction and communication were identified through the analysis: 1) commenting and voting on existing ideas, 2) placing icons, and 3) drawing sketches. Since moderation is often conducted manually, this impacts which features can be implemented. We also found that city staff was generally interested in using AI for several purposes, such as lowering their moderation load and supporting citizens in formulating ideas.

CCS Concepts

• **Human-centered computing → Empirical studies in collaborative and social computing**.

Keywords

co-design, e-participation, qualitative study, 3D visualization, sociotechnical systems

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1 Introduction

Citizen participation is an important way in a representative democracy to understand the desires, needs, and opinions of citizens in the policy-making process [32, 71, 85]. By using the Internet on various devices, such as computers and smartphones, citizens can participate digitally and asynchronously. Including citizens in decision-making processes through information and communication technology (ICT) is referred to as e-participation [66, 134].

E-participation has been described as a way to make decision-making processes transparent and build trust between the municipality and citizens, as well as improve the democratic responsiveness of the political system [51, 52, 71, 128]. However, empirical evidence shows that e-participation is not always successful in achieving these goals [8]. Reasons include low participation rates and the perception that there is a lack of feedback and opacity in processes [17, 50, 73]. Design considerations are another aspect that could impact the success of e-participation [76, 114], as the implemented features can strengthen some modes of participation and make others impossible [107, 129].

We are currently developing an e-participation tool in collaboration with two German municipalities. Although there are many decisions citizens can be involved in, we focus specifically on participatory urban planning. The aim of developing a new tool based on human-centered design is to foster cooperation among citizens and other stakeholders to increase the acceptance of planning processes and especially improve the participation of previously underrepresented groups [82]. To develop a tool based on user needs and preferences, we conducted co-design workshops with citizens, city staff, and experts on participation [95]. Due to our interest in the views of groups underrepresented in e-participation [28], our participants included senior citizens and immigrants. This paper focuses on the generative phase of the design process, to produce ideas, insights, and concepts, and to discover unanticipated user needs [95]. We aim to answer the following research questions from the perspective of our participants:

(1) What information do citizens and city staff need, for which purpose do they need it, and how should this information be provided?
(2) What features for input and communication should be implemented?

From our findings, we derive general recommendations and considerations on the design of e-participation tools and describe a first prototype.

2 Background and Related Work

Citizen participation aims to involve citizens in decision-making processes [32] to improve the democratic responsiveness of the political system, strengthen the participation of citizens between elections, and build trust [13, 25, 128]. By integrating relevant information provided by citizens and through public deliberation,

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citizen participation processes can contribute to the acceptance of government decisions, create transparency, impart democratic skills and attitudes, and contribute to social integration [38, 68, 71]. Participation can take place online and offline [77]. E-participation generally refers to participation conducted through the Internet with the use of computers, smartphones, and other mobile devices [134]. We follow the definition of Macintosh [66] who describes e-participation as using ICT for online consultation and dialogue between government and citizens. E-participation can lower barriers and reach a wider audience by offering participation channels that can be accessed independently of location or time [24, 32].

There is not only *one* way of involving citizens. In the ladder of participation proposed by Arnstein [7], the levels of participation correspond to three general forms: nonparticipation, degrees of tokenism, and degrees of citizen power. The OECD describes increasing levels of citizen involvement and influence on policy-making ranging from information to consultation to active participation [78]. Due to its continued relevance and popularity, we will focus on Arnstein's typology. While these are only two examples of categorizations of citizen participation, they have a common focus on the *actual* power citizens have in implementing ideas.

In Germany, the ultimate decision-making authority remains with the representatives authorized by the election [10]. This means that citizen participation remains consultative and therefore has to be distinguished from instruments of direct democracy, which do result in binding decisions that bypass the elected representatives [51]. Arnstein [7, p.217] argues that if citizens are only consulted (part of tokenism in the typology), they "may indeed hear and be heard" but that their views are not necessarily heeded, and that there is no guarantee of being able to change the status quo. In addition, most procedures for citizen participation are initiated top-down by the local authority [70, 100].

Royo et al. [88] found that participation processes in both Germany and other countries, such as Spain, are often used to comply with legal requirements or to increase the perceived legitimacy of decisions, but that citizen input may not be used sufficiently to enrich the decision-making process [88]. Other research found that both citizens and local politicians are in favor of expanding participation opportunities [34, 42, 108]. We argue that involving citizens in the decision-making process through consultation can enrich the decision-making process of representatives, especially if the e-participation tool used is well-designed. In the following sections, we will explore the findings of previous research on problems that can impact e-participation processes.

2.1 E-participation from a Sociotechnical Systems Perspective

Although many e-participation tools have been developed and are currently in use, there have been reports of failures of such tools, which is why previous research has called for a sociotechnical approach to e-participation [114]. A sociotechnical system consists of interdependent but separate technical and social subsystems which are interacting with each other [12, 43, 61]. The social subsystem refers to factors such as people and their skills, values and attitudes, and organizational structures [12]. Sociotechnical system theory implies that 1) system performance depends on the interaction between the social and technical subsystems, and 2) optimizing only one of these will decrease system performance [126]. Although originally applied in contexts such as coal mining and factory work [117, 119, 129] it has been translated to digital work processes, shifting the focus beyond an organization to a larger ecosystem. An example is the digital sociotechnical systems approach by Winby and Mohrman [129] which addresses the need to understand the complexity of digitally enabled work systems while acknowledging diverse interests and the need to enable interactions between technology, individuals, and other actors in the larger ecosystem.

Susha and Grönlund [107] argue that the development practice of tools is often focused on delivering a technical solution and neglects to properly consider social and organizational factors, such as resources and capabilities of future users. The authors describe a tool with limited features for online deliberation that led users to take their discussions offline. User expectations and the purpose of the system were not considered sufficiently during the development phase; joint optimization did not take place [67].

Understanding *how* citizens want to participate, especially how they want to contribute their ideas and opinions and discuss them with the city administration and other citizens is important to avoid under-use and lack of impact of a tool [114]. Ways of communicating with other citizens and the municipal government offered in previous e-participation tools are writing comments and giving likes/dislikes to contributions [44, 60]. Some also offer the possibility of uploading pictures, videos, and voice messages or allow sharing comments through social media or in private chats [39, 113]. Visual information can support e-participation by allowing the placement of markers, icons, or other objects on a map or in a 3D model, generating pictures, and drawing on a map [2, 59, 64]. Displaying different plans developed by urban planners is another possibility [37, 115]. These ways of sharing information and communicating with others mirror possibilities of interaction during an in-person workshop [82]. Some projects also use gamification in the form of leaderboards, badges, and other game design elements to raise levels of engagement [111]. Due to the importance of communication in e-participation, RQ2 focuses on how users want to interact with e-participation tools and in what way they want to communicate with other users of the tool.

Adoption and use of e-participation tools depend on social factors like trust [99]. Santamaría-Philco and Wimmer [96] regard transparency as a factor that influences trust. Transparency can be improved by providing citizens with important information such as which tasks need to be done (e.g. sharing ideas or only rating ideas) and the scope of participation (e.g. budget restrictions). Moreover, information on decision-making processes is important, e.g. feedback on how citizen input is processed and which ideas or preferences are ultimately implemented. If such information is not shared and the information flow remains unilateral, participation might be perceived as opaque and lead to lower satisfaction with e-participation [50, 55]. However, the government is subject to budget restrictions, and a shortage of staff can lead to a lack of adequate processing of feedback from users [107].

Venkatesh et al. [122] found that while the fact *that* the information is shared is important, it is also important *how* it is shared [90]. Afzalan et al. [1] argue that information generated by citizens through participation tools should be credible (accurate arguments), salient (relevant information), and legitimate (information based on diverse beliefs and values of stakeholders) to be effective in policy-making. Urban planners can benefit from well-structured input when developing ideas, and well-structured input might have a greater chance of being considered or even implemented by the administration and politicians.

In participatory urban planning, e-participation has often been supported through visual maps [56] and in more recent years 3D city models, Virtual Reality (VR), and Augmented Reality (AR) [18, 97, 101]. Visualizations can help citizens understand the impacts of new urban planning projects. Conveying information visually can facilitate dialogue between citizens and municipalities [18, 59, 127]. Recent advances in Geographic Information Systems (GIS) which allow for 3D modeling, have led to e-participation tools that use 3D models ranging from very abstract "block models" without roofs (e.g. [9, 16]), to models with an abstract roof structure [131], to photorealistic models (e.g. [11, 79, 103, 133]). 3D models can display additional details about an area, such as weather conditions (precipitation, wind speed, and more), traffic conditions [18, 26], or information on particular buildings, such as empty shops [133]. Common functions in 3D models include zooming, rotating, and flying along a route in the model, making an area more tangible than through a map [130]. Due to the importance of having access to information for both citizens and city staff, we aim to find out how information should be provided and whether visualizations can support information sharing (RQ1).

2.2 Including Stakeholders in Public Service Development

There are previous studies that included stakeholders in the development of participatory urban planning tools. Kwon et al. [59] describe the results of three community design workshops that focused on an interactive, photorealistic 3D visualization to support participatory neighborhood planning. The authors found that the 3D visualization enabled users to situate themselves in the space, supported them in imagining usage scenarios, and guided them in placing objects. While previous studies mainly focused on photorealistic 3D models for participation (e.g. [59, 79, 130]), more abstract models have the advantage of reducing loading times and needing fewer data which is especially advantageous for mobile use [130, 132]. However, too little detail could potentially have a negative impact in the context of participatory urban planning [18]. Our approach to investigating this issue builds on previous research by inquiring about the suitable level of visual realism for this context.

When viewing e-participation from a sociotechnical perspective, it is important to understand the needs of users in this specific context which can be achieved by involving them during the development phase through co-design [80]. Vargas et al. [121] understand co-design as an "active collaboration between stakeholders in designing solutions to a prespecified problem", in our case this problem is the design of a tool for e-participation. Our research adds to previous co-design studies for public services, including those of Kawakami et al. [53], Fröhlich et al. [30], Forbes and De Paoli [29], and Jannack et al. [48]. Similarly to Kawakami et al. [53] and Fröhlich et al. [30], we include those directly impacted by the introduction of tools by taking into account the context and skills of different stakeholders to design a tool according to their needs and preferences. While Stelzle et al. [106] describe a co-design process for their e-participation platform, the authors focus on user stories instead of a qualitative analysis of their workshop data. We add to the existing body of literature by presenting a thematic analysis of our workshops conducted with different stakeholders and discussing our findings through the lens of sociotechnical systems.

3 Method

We expect the acceptance of e-participation tools to be higher if accompanied by a well-designed participation process based on human-centered design (HCD) [98]. HCD focuses on the "use of techniques which communicate, interact, empathize and stimulate the people involved, obtaining an understanding of their needs, desires and experiences" [36, p.610]. ISO9241-210 provides requirements and recommendations for HCD in the context of computer-based interactive systems [104]. Among the recommendations are the "explicit understanding of users, tasks and environments" and "the involvement of users throughout design and development" [36, p.608]. HCD is viewed as an sociotechnical systems design approach by Baxter and Sommerville [12] due to the focus on an explicit understanding of users, tasks, and the use context.

Similarly to previous work following a sociotechnical systems approach (e.g. [80, 129]), we conducted co-design workshops to consult the expertise of different groups in an early design phase to derive design criteria and ideas for our tool [93–95]. We conducted five co-design workshops in Wuppertal, Germany, to explore our research questions. The workshops took place from January to May 2024. Our workshop activities were inspired by two categories corresponding to the research questions: 1) information (both textual and visual), and 2) interaction and communication. Two of the five workshops were conducted with citizens, one was conducted with experts on participation, and two were conducted with employees of the city administration (see Table 1).

During the workshop, we used three visualizations (see Figure 1). As mentioned in Subsection 2.1, 3D city models (both abstract and photorealistic) have been used in citizen participation, and 2D maps are well-known through map applications. Building on previous research (see Subsection 2.2), we wanted to find out perceived advantages and disadvantages for each visualizations. In particular, we were interested in what type of visual information should support textual information to foster transparency.

3.1 Participants

Previous research found that most users of e-participation tools are young adults, as well as middle-aged, highly-educated, and technology-savvy citizens [28]. Design influences the degree to which different groups of citizens can use a tool, and accessibility is an important consideration [31, 72]. We aim to improve the accessibility of participation processes for groups currently underrepresented in e-participation [28, 67, 116] and to improve participation for those who are facing barriers [49, 65, 120]. Based on these aims, our workshop participants included members of a group of senior citizens and members of migrant associations.

	Citizens	Administration	Experts
Number of Workshops	2	2	1
Number of Participants	12 (Workshop MA), 5 (Workshop SC)	11	11
Location	in their regular meeting rooms	in the city hall	online through Zoom
Date	April 16 (MA) and May 16 (SC), 2024	January 30 and March 14, 2024	April 19, 2024
Material	pinboards, moderation cards, stickers, posters, presentations, screen recordings, screenshots, mock-ups on iPads and Android smartphones	pinboards, moderation cards, stickers, textual descriptions and screenshots of features in existing tools	sticky notes, digital stickers, screenshots of existing tools and our mock-ups in a ConceptBoard

Table 1: Overview of key information on the five workshops (MA refers to the workshop with migrant associations and SC refers to the workshop with senior citizens)

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Figure 1: Screenshots from the three visualizations that participants could switch between in the mock-up. The left screen shows a 2D map, the center screen shows an abstract 3D model and the right screen shows a simulation of a photo-realistic 3D model through a photo. It was possible to zoom in and out and give a like for an example idea. The mock-up was developed by the company viadukt GmbH.

These associations form a network that comes together in regular meetings coordinated by the city administration. These meetings are intended to foster exchange, cooperation, and participation and strengthen the voice of migrant communities in Wuppertal. The associations represent people with migration experience (both immigrants and those temporarily moving to Germany) from various cultural backgrounds.

Participants for the workshops with citizens were recruited through existing networks of the municipalities. Table 2 provides information on participant demographics from the workshops with citizens. Most participants of workshop MA have foreign citizenship, but have been living in Germany for decades (since the 1980s and 1990s), or have obtained German citizenship; only one person immigrated in 2014 (first-generation immigrants). One of the younger participants was a second-generation immigrant. One participant did not answer the questions on citizenship and age at migration. Despite a generally high interest in politics among the participants, the amount of participation in previous participation formats ranged from zero to three. Similarly, the participants of workshop SC have a strong or very strong interest in politics, but the number of times they participated in previous citizen participation formats ranged from zero to four times. By involving these groups, we hope to gain a better understanding of the interests of potential future users and their level of comfort when using digital tools [69].

We did not collect demographic data from the experts or city staff. Three employees and one trainee from the administration's team for citizen participation took part in the first workshop. Two employees and one trainee from the team also participated in the second workshop. In addition, seven employees from departments focusing on urban construction, integrated urban development and planning, new construction, and overall transportation attended. All employees of the administration had previous experience with citizen participation.

Our workshop with experts was comprised of 11 participants who all have an academic background and are working at different universities and research institutes in Germany. Expert in our case was defined as a researcher who has previous knowledge in the field of e-participation or visualization of cities through publications or projects [19, 27]. Participants were recruited via e-mail. It was important to us to include the opinions of experts in the field to avoid making common mistakes and receive feedback on our ideas.

3.2 Workshops with citizens

The workshops took place in April and May 2024 during regular meetings of citizen associations with which we were in contact. In the following, workshop MA will refer to the workshop with migrant associations, and workshop SC will refer to the workshop with senior citizens. Both workshops started with an introduction of the facilitators and the research project. We asked participants to share their knowledge of citizen participation (including previous experiences). We then provided definitions for the terms citizen participation and direct democracy to have a common understanding of the topic. For the activities, we chose a scenario-based approach in which we asked participants to imagine that the planning process for a park will be conducted through our tool [87]. This park had been the subject of an e-participation format in the past (shown as a screen recording). The participants were given iPads and smartphones to interact with the mock-up (see Figure 1). It was possible to switch between visualizations of the park which correspond to a 2D map, a 3D block model, and a 3D photorealistic model. We wanted to include a mock-up in our workshops to act as a basis for discussion, in the form of an artifact that is unfinished and open to experimentation but still allows participants to experience a potential e-participation process [105]. It was possible to zoom in and out, click on icons that represented ideas, and give a thumbs up for these ideas. By presenting different options for the visualization of the park, we wanted to discuss which was the preferred one for participatory urban planning to find out what visual information citizens need and how this information should be provided (part of RQ1). After participants had time to test the mockup and share their initial impressions, we asked which visualization was their favorite and why. We then asked participants to vote on a few criteria: which visualization was the most realistic, the most suited for orientation, the easiest to learn to use, the most experienceable (in the sense of getting a good overview of the area), the most aesthetic, and the most fun to use. The votes served as a basis for discussion among the participants (see Figure 2). Since Wuppertal is a hilly city, we were particularly interested in whether information about the terrain would be needed and whether vegetation such as trees should be depicted realistically to contribute ideas to the urban planning process.

In the next part of the workshop, we continued to focus on information needs (such as the content of information and its presentation) and potential features for input and communication (to answer RQ2). In an open brainstorming part, we collected participants' ideas and presented our own to discuss (such as placing 3D objects or giving a thumbs up). The next activity was a discussion in a small group (two to three per group) for which we provided example questions such as "What do you need to be able to participate meaningfully? How would you like to contribute your opinions?". We asked the participants to use the example of the park and the mock-up as a basis for discussion and to write their results on moderation cards. The moderation cards were pinned to a pinboard, clustered by topic, and each idea was discussed with all participants who commented on the advantages and disadvantages of each idea. At the end, participants were able to share their opinions on other topics and ask questions.

3.3 Workshops with city staff

The two workshops with staff from the municipal government took place in January and March 2024. The focus of the first workshop was the discussion of an existing e-participation tool to derive ideas for the development of our tool. Criteria for successful citizen participation as determined by the city council served as the basis for ideation. Participants were also able to openly suggest new ideas or comment on our ideas. The second workshop focused on the backend and aimed to identify requirements and ideas for features that city staff can use to initiate, manage, and evaluate participation results. The activities of both workshops included brainstorming in groups, moderated discussions among all participants, and prioritization of the discussed ideas. Existing 3D models were presented on a smartboard to discuss their advantages and disadvantages.

3.4 Expert Workshop

The workshop with experts on participation took place on Zoom on April 19, 2024. A Conceptboard[1] was used to support dialogue during the activities by allowing participants to write down their ideas and present the visual material we provided. The participants were asked to join two smaller groups based on their personal expertise and interests. Group one was facilitated by the second author and focused on input and communication (RQ2). Group two was facilitated by the first author and focused on (visual) information conveyed by maps and 3D models (RQ1). Group one started with an open brainstorming part to collect possible modes of participation in a new e-participation tool. The next activity used a table with ideas for information sharing and ways of input and communication that we prepared beforehand. The participants were asked to discuss advantages, disadvantages, and additional considerations for each idea. Results of the brainstorming were also added to the table to discuss. Participants were then asked to prioritize the three ideas which they would like to see implemented the most by pulling a symbol to those ideas.

In group two, participants were first asked about their previous experience (such as whether they had used 2D or 3D models before and if yes, in what context). Then, screenshots from other citizen participation projects in Germany that use 3D models were discussed. The participants were asked to argue whether they see advantages to using 3D visualizations over "classic" 2D maps in e-participation. The participants were then presented with screenshots of the three mock-ups (see Figure 1), and asked to discuss advantages and disadvantages of each visualization. To indicate their favorite visualization, the participants were asked to drag a start to the corresponding screenshot. After closing the break-out rooms, both groups gave a brief overview of the results of their activities which were then discussed with all participants. At the end, there was time for further discussions and comments.

3.5 Ethics

Before the workshops started, informed consent was obtained by giving the participants a form that included information on the aim of the study, anonymization procedures, possibilities of requesting data deletion, storage duration, and other topics. Participants were given time to read the forms and ask questions before signing. The workshops with citizens and experts were recorded and transcribed for analysis, and during the workshops with city staff, we took notes. Data were anonymized and stored in a GDPR-conform way. For us, it is important to keep our participants in the loop by providing them with updates on developments within the project. This is why we sent summaries of the workshops to participants and provided them with ways to stay informed and involved, e.g. by providing a link to our project website.

3.6 Data Analysis

To analyze our data, we conducted a hybrid approach to thematic analysis after Boyatzis [21] in which we used both inductive and deductive coding. Our review of previous literature (see Section 2) and our research questions informed the coding frame that was developed a priori. We agreed on the following code categories with pre-defined sub-codes for our deductive coding: 1) information, and 2) interaction and communication [45]. Transcriptions of audio recordings, photos, notes, and screenshots from the workshop activities served as the basis for the coding conducted in MAXQDA. Participants from the citizen workshop are referred to

[1]https://app.conceptboard.com

	Workshop with migrant associations	Workshop with senior citizens
Age range	30-57	69-80
Average age/Median Age	48.1/51	75/77
Self-identified gender	8 women, 3 men	3 women, 2 men
Education	university degree (5), vocational training (3), high-school diploma (2), no answer (1)	vocational training (3), secondary school leaving certificate (1), no answer (1)
Languages spoken at home (more than one possible)	Turkish (5), Arabic (1), Tamazight (1), Russian (1), German (8), no answer(1)	German (5)
Citizen participation experience	previous participation ranged from zero to three times	previous participation ranged from zero to four times

Table 2: Comparison of participant demographics from the workshops with citizens

by the pseudonyms P1 to P17, and the experts E1 to E11. Verbatim quotes by city staff taken from notes are referred to from A1 to A11. All material was in German and coded by the first author, and the second author read all transcripts and coded some material to compare coding results and discuss the codes. After coding was done, the first author constructed a table summarizing the opinions of the participant groups and highlighted the similarities and differences between them. The themes derived from this comparison of similarities and differences were discussed among all authors. The final codes can be found in the Appendix. To present the findings, we translated quotes into English.

4 Findings

4.1 Fostering Transparency Through Information: Where, What and How?

From our analysis, we found that there are three important considerations when it comes to the provision of information. The first is that it needs to be clear *where* information can be found. Participants from both workshops with citizens kept mentioning how important it is to stay up-to-date and receive feedback on one's ideas. However, participants from workshop SC told us that they were unaware of how to receive news and updates. During the workshop, we were asked if we could help them subscribe to a newsletter and what information the newsletter contains. Older participants were very interested in analog ways of staying informed, such as displaying plans and flyers in public spaces, as well as publishing information in local newspapers.

The second consideration is the *what*, i.e. the content of the information. It was important for participants to know whether citizens' ideas (not necessarily their own) are included in decision-making processes and to what extent. Participants argued that while not all ideas can be taken into account, they would like to know what ideas qualify as useful and how many can be considered. They also mentioned the importance of interim reports, final reports, personalized feedback, and notifications. Participants wanted to be notified when someone interacts with their idea by commenting on it or liking it, and would ideally like to get personalized feedback in the form of an assessment of the feasibility of their idea. Some participants not only wished for feedback from city staff but also emphasized the importance of seeing other citizens' ideas and comments which can inspire new ideas or lead to discussions:

> Those are some things that you want to see: what did the other person say? Maybe even: What do they think about my idea? (P12)

P5 had a similar opinion as P12 and said that she would like to see and respond to other users' ideas to see whether something is missing or something she disagrees with entirely was proposed. P2 on the other hand wanted to avoid being influenced:

> When I have a completely empty area, then I have the possibility of letting my creativity run free. [If I see] others' ideas, I am biased and might not be able to contribute ideas, that I would probably contribute. (P2)

During the workshop with the administration, the limited resources for personalized feedback were discussed. It was suggested that Artificial Intelligence (AI) could give direct feedback to citizens while using the tool. This would include comments on whether their idea can be implemented or not, and help with navigating the tool.

Another aspect of more personalized notifications was the wish of some participants from workshop SC to receive only notifications based on their district of residence, while others thought that citizens should be interested in all districts. P13 commented on the reasons:

> If I had relatives [in that district] or people I know, then I would be interested. But when I have no connection, then I don't know if I should get involved. Not really any of my business. (P13)

A1 argued that she believes it is unnecessary to offer personalized notifications, as there is a very small number of e-participation formats per year, and that this could lead to more work for the administration. In addition, there was a discussion on whether staying up-to-date on everything that happens in the city of residence should be preferred, as it could otherwise lead to filter bubbles, in which only certain information gets transmitted.

Surprisingly for us, participants also wanted to discuss the registration process in both workshops and described it as too complicated due to the high amount of information that needs to be given as input on the currently used e-participation website in Wuppertal.

An idea by city staff to offer more information was to publish drafts from urban planners with background information, such as costs, sustainability, and consequences of implementation. A concern in this debate was whether publishing pre-set drafts can

Present	Future
Too much (complicated) text	More visual information (pictures, photos, videos), simple language, ideally different languages
Maps too cluttered (overlapping icons)	Filtering of contributions and clustering for different topics, offering the possibility of viewing an empty map
Uncertainty of how to receive information	More analog advertising, make newsletter subscription more salient
Complicated registration process	Easier navigation of the tool with less information needed during the registration process
Perception of missing feedback or feedback presented in a form that is hard to understand	Progress pictures, tables with pros and cons for ideas, online consultation hour

Table 3: Citizens' criticism of current information presentation and dissemination, and wishes for future improvements

appear limiting to citizens (the scope in which they can contribute appears too narrow). Another open question was how it can be avoided that citizens feel overwhelmed by the presented information. It was suggested that for complex use cases, the tool should be used during an in-person event, in which trained staff can support the use or the tools can be used together.

The third consideration is the *how*, i.e. the presentation of the information. For an overview of all discussed points regarding the presentation, see Table 3. Participants from citizen workshop MA told us that while they were aware that interim reports are available, they found the provided information to be too text-intensive and expressed a preference for more visual information and simple language, as well as better navigation on the website, such as the possibility of improved filtering of information.

During the workshop with the administration, it was also suggested that to ensure high-quality contributions, very motivated users should be able to filter and sort information in different categories (such as ideas on improving playgrounds, advantages to a certain idea etc.). Further information on how to use the tool (e.g. a longer tutorial), and information on the planning process (e.g. further information on the available budget) in a longer, more extensive form should also be available to these highly motivated individuals. This information should be presented in both visual and written form. Answers to the following questions were deemed to be important to citizens according to the administration: Why should citizens participate? How can they participate? Is there a personal impact on citizens? What is the scope of participation? What participation processes are currently available?

Participants of workshop MA suggested short videos (ideally in different languages) as a way to convey the most important information for an e-participation format. P2 referred to a video she watched during an on-site event:

> There should be a video on the platform so that you can get some information first. Very simple, very easy, very uncomplicated. A way to learn about the most important information. (P2)

Visualizing the area of participation in an urban planning process can support citizens in their contributions. The abstract 3D model was the most popular among participants. During workshop SC and MA, participants described the abstract model as giving a good overview of the area, being non-distracting, more focused, and better suited for navigating the area. However, participants also mentioned disadvantages: the differences between hills, grassy areas, forests, and parks were not clear enough in our mock-up, which could lead to being unable to identify unsuitable areas to place objects. A notification should be displayed if an object cannot be placed. Some participants preferred the photorealistic model and described it as supporting the recognition of the area and the development of ideas. P14 argued that while the block model is too abstract to recognize the area properly, some knowledge of the location should be a prerequisite to participation:

> I prefer this one because it is a realistic visualization. The other one is too abstract. I cannot imagine the park, when I see this green blob on the map, you know? Well, if I am familiar with the area, then I can imagine something, of course. But I want to add, that participation is only called for if you are familiar with [the area], if you have some kind of connection to it. (P14)

Working with the abstract model was perceived as easier by some participants, as it was seen as "less chaotic", yet still realistic enough. In both citizen workshops, it was mentioned there should be information on public transport and traffic, such as speed limits, one-way streets, and other information related to the safety of children.

The city staff was also interested in ways to visualize information or simulations through a 3D model, such as how many people pass by an area, what shops are in the area, available parking spots, water retention, and noise pollution. To experience the area even better, it was suggested that virtually flying over the area from a drone view should be possible. Other ideas included that it should be possible to show different stages of development and changes over time. In addition, there was an interest in visualizing negative scenarios (what could happen if nothing changes in this area?), possibly through AI.

During the expert workshop, E6 argued that the effort to gather data needed for a highly realistic model is unnecessary and that long loading times can lead to users getting frustrated. A discussed possible alternative to a photorealistic model was adding a layer to an abstract model that visualizes the terrain. This layer would not be shown by default to save loading time and technical resources. Based on the use case, information such as alternative routes and other traffic information could be visualized through differently colored lines. The area of participation could be marked with lines

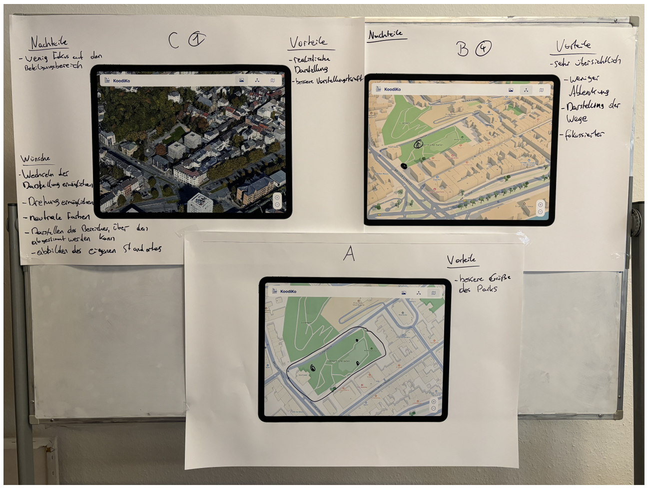

Figure 2: Poster from citizen workshop SC with disadvantages and advantages for each model, and wishes for improvement. Translation: [Left poster] "Disadvantages: low focus on the area of participation; Advantages: realistic visualization, better recall; Wishes: changing between visualizations, rotation, neutral colors, marking the area of participation, show own location", [Right poster] "Advantages: very good overview, less distraction, routes are shown, more focused"; [Center poster] "Advantages: better size of the park"

as well. We also discussed whether there is any advantage to 3D models over 2D models. Arguments for 3D models were that new buildings can be visualized with accurate height differences to the surrounding area and that the visualization of wind tunnels, airflow, and shade can help understand the consequences on the environment.

4.2 Offering different choices on how to participate

In all workshops, we discussed how to enable citizens to interact and communicate with the tool, each other, and the government according to their needs. We identified three preferences for interaction and communication: 1) commenting and voting on existing ideas (from other citizens or the government), 2) placing icons in addition to commenting and voting, and 3) sketching ideas in addition to placing icons and commenting and voting.

The participants of citizen workshop MA were interested in placing icons, but also having the option to sketch ideas. This was seen as a good way to express ideas beyond available icons, e.g. drawing an entrance gate to the park. Experts cautioned that while sketching is a good way to capture ideas beyond pre-set icons, there could be an overestimation of the scope of participation.

The participants of workshop SC were mostly interested in placing icons and less interested in sketching. Giving a thumbs up or down was seen as a good, low-effort, and easy way of expressing an opinion and participating. Overall, the participants reflected on their own need for feedback at various times. They contrasted how interaction and communication that goes beyond pre-set options can burden the city staff who have to moderate and evaluate the submitted ideas which may be "excessively long" (P4) if not restricted. Uploading photos and videos was rejected, as this would become "too negative" (P1) through a high number of posts and comments, and this was seen as more suitable for social media than e-participation. The possibility of placing 3D objects was also rejected, as this was seen as "unnecessary" (P1) or as "attractive but not feasible" (P3) due to the moderation load on city staff. However, it was mentioned that such features could be suitable for students or in a school context, as this would allow a more playful way to participate and get to know e-participation.

P6 expressed her wish for a planning tool similar to those used by architects. In the workshop with city staff, such a planning tool was also discussed and it was proposed that users could receive comments and ratings of their drafts in the tool. The idea was that this could be connected with a proximity chat (inspired by WonderMe or GatherTown), to foster communication among citizens. Apart from making their own drafts, citizens should have the possibility of seeing drafts from urban planners next to the status quo and commenting on them. The idea of placing objects in an area of the 3D model to make drafts was also well-received, and it was suggested that there could be objects like trees, benches, and garbage bins available, or that users could also upload objects. However, it was mentioned that while users might find such ways of participatory urban planning fun, it could overwhelm them, both due to an overestimation of the scope of participation and a lack of guidance. The city staff was thus particularly interested in using such a tool during on-site workshops with citizens, politicians, or other groups, and being able to interact with the 3D model by drawing new routes, showing different perspectives on areas, and asking for opinions.

During the expert workshop, there was also a discussion on the feasibility of such a planning tool and the conclusions were similar to the ones from the workshop with the city staff. A proximity chat was seen as a possibility to foster discussion and dialogue between citizens based on spatial and thematic closeness. The collaborative development of drafts and peer pressure were viewed as potentially reducing negative behavior that could occur in direct or private messages. Several caveats similar to those by city staff were mentioned, including a potentially high amount of unfinished drafts due to the complexity of the task and the time consumption. Experts agreed with city staff that legal or budgetary restrictions could lead to frustration and that it would be a feature better used during in-person workshops.

During workshop MA, it was mentioned that there could be online consultation hours. The city staff also suggested such a feature. During such a consultation hour, citizens would be able to ask questions regarding the planning process and receive feedback for ideas. This could be done via video call. However, city staff mentioned that receiving feedback that an idea is not feasible might lead to feeling demotivated and inhibited creativity.

The city staff said that they would be open to receiving photos and other attached files. Experts argued that the possibility of uploading content beyond comments can "extremely expand the creative space" (E3) but that it needs an open mind from the city staff, and the length needs to be restricted to avoid overburdening the administration. In addition, content needs to be moderated to avoid publishing "nonsense" (E3) and "inappropriate" (E3) contributions and there needs to be knowledge of potential issues with data protection.

Crowd mapping is already popular with the city staff since citizens can comment on strengths, weaknesses, or new ideas for the urban space and they would like to continue using crowd mapping in future tools. Currently, every comment is manually moderated and published after a screening, and city staff expressed the wish for automatic text processing for comments and clustering by topics.

5 Discussion and Future Work

From the data analysis, we found that while there are issues that all participants agreed on, there are also differing needs, especially when it comes to information citizens have to provide for the municipality and vice versa (see Table 4). From our analysis, we found that the main problems of why citizens' needs are currently not met are due to limitations of current e-participation tools and resources, not an unwillingness to meet these needs by city staff.

5.1 Recommendations Regarding Information (RQ1)

The different participant groups generally agreed that information should be short and predominantly visual. In addition, there should be filters to avoid being overwhelmed by seeing all available information at once. Similar to previous research by Tscharn et al. [118], we found that there are currently existing ways to stay informed on the progress of participation formats, but some citizens are unaware of them. Based on our findings, we derive some general recommendations for e-participation tools:

- Share information multi-modally: We recommend that e-participation tools should enable users to share and receive information in a multi-modal way, through the possibility of not only adding text, but also photos, pictures, and videos.
- Support citizens with AI: Previous research has focused on AI for political participation, e.g. in the form of an intelligent virtual assistant [22] or conversational agents/chatbots [14, 102]. We recommend that municipalities and universities collaborate to conduct studies on how such features influence user experience and whether they are perceived as helpful by citizens and city staff. On the technical level, there is the need to detect and mitigate "hallucinations" (plausible, yet nonfactual content) in output of large language models (LLMs) [46, 125]. Strategies to mitigate hallucinations include retrieval-augmented generation [92]. The implementation of AI-based features beyond the scientific context has to be carefully considered from a legal perspective, since LLM providers currently fall into a gap in existing EU legal frameworks regarding the duty to minimize careless speech [125].
- Use an abstract visualization: Based on our findings, we propose that an abstract 3D model has the appropriate amount of detail for a user familiar with an area to participate. This is in line with research conducted by Jaalama et al. [47] who found that perception of visualizations is not only dependent on the level of realism but also on prior information available to the user or a prior visit. We recommend adding additional, more detailed information, such as photos or videos of the area, or an additional layer visualizing the terrain. Future work should further explore or evaluate the use of abstract 3D models in participatory urban planning.
- Offer different ways of visualizing information: Visual information such as the number of parking lots, mobility options, pedestrian flows, and traffic conditions can support deliberation by making current circumstances easier to understand and serve as a basis for discussions. Other ways to improve information processing include computer-supported argumentation visualization tools [15]. When implementing such approaches, there has to be additional training for city staff on how to add information to the tool and tutorials for citizens on how to use the features.
- Simplify the registration process: While the registration process was not part of our workshop activities, participants of the citizen workshops mentioned their difficulties with it on their own accord. The citizens' need for an easy, accessible registration process or even no registration clashes with the need for security in the form of information about a user to prevent abuse [62, 81, 112]. In the particular case of Wuppertal, citizens described that the information asked during the registration process went beyond providing an e-mail address and setting a password, and asked for further demographic information. We suggest asking for demographic information at a later point, for example on a user profile page. Experts and city staff agreed that registration is necessary but that indicating a pseudonym instead of the name should be sufficient. An easily accessible possibility to prevent bots from spamming while offering a low threshold for participation would be to distribute QR Codes to specific target groups [63]. This should not replace a regular registration process for other users but could be offered in parallel. Other approaches include setting up feedback pillars that allow citizens to press buttons to answer in areas that are part of participation processes [41].

5.2 Considerations Regarding Input and Communication (RQ2)

As mentioned in Section 4.2, we identified three preferred modes of interaction and communication: 1) commenting and voting on existing ideas, 2) placing icons, and 3) sketching ideas. We interpret that some participants value creativity and having the choice of how to convey their ideas, while other participants prefer clear instructions and less creative ways of contributing, such as giving thumbs up for an existing idea. How citizens can participate should also depend on the goal of the participation process. The aforementioned modes of participation are suitable for questions in the early phases of the redesign of an area. In later phases, in which concrete drafts are already available, commenting and voting might be more suitable. In addition, there was a clear difference in preferences between the younger workshop participants in workshop MA and the senior citizens in workshop SC. We suggest that designers and developers should keep these considerations in mind:

Citizens need...	The administration needs...
Both pre-defined and creative ways of creating contributions	Contributions that are within the scope of participation and in a suitable format
A civil, constructive environment for discussion both with other citizens and the administration	Ways of enabling discussions that match the existing resources for moderation
A simple, uncomplicated registration process or no registration	At least basic information to prevent abuse

Table 4: Three examples of conflicting needs between the citizens and the administration

- Available resources for moderation: Abuse can occur with textual or visual contributions, especially if combined with a high degree of anonymity [23]. Moderating content requires financial resources and time that not all municipalities have, and often contributions are screened manually. Machine Learning (ML) approaches could be used to support the analysis of citizens' opinions, such as topic modeling to group them, and sentiment analysis to determine whether comments express a negative, neutral, or positive sentiment [3, 6, 33, 40, 73]. Another way citizens' input can be analyzed is stance detection (predicting the writers' stance on a subject of interest) [4]. An important consideration when using these approaches is their accuracy, which depends on the model used, the language (most models are developed for English), and the quality of the data [84]. In addition, data protection laws must be considered when compiling a training data set [58].
- Choice of participation modes: We found that placing 3D objects was seen as unnecessary and a feature that would be better suited for specific groups, such as students. A tool that is aimed at a broad group of people should offer a mix of low-effort and creative ways to participate. Offering only pre-defined icons and objects to place can reduce the moderation load, but leads to a restriction on the creativity of participants and impacts autonomy regarding decision freedom, i.e. how to interact with the tool [89, 91]. A way to allow some level of creativity but not completely free input could be to offer generative AI features, which allow citizens to visualize their ideas through prompts [124]. Preussner et al. [83] conducted a study in which they asked participants to use a generative AI tool to submit ideas for urban development by taking pictures and manipulating them. Among the identified challenges was that some participants struggled with the wording of prompts and thus did not achieve their desired results. A way to mitigate this problem might be to give participants some information on how to formulate prompts prior to the study [35, 54].
- Specific features exclusively for in-person workshops: The planning tool discussed in the findings requires in-depth knowledge about legal and financial conditions and was perceived as too complex to make accessible without guidance. City staff and experts suggested that this tool can be a great addition to in-person workshops where direct feedback can be given and the administration and citizens can work on plans together.
- Content and type of contributions: Some see e-participation as a way to significantly lower the cost of involving many rather than fewer citizens, and to include an almost limitless number of participants [28, 134]. However, we argue that this depends on the system design. If citizens can freely produce content, there is the issue of considering the feasibility of ideas and the city staff's ability to translate the desires of citizens into formal legal policy [86]. It could be beneficial to provide citizens with instructions and examples of contributions that are considered helpful by city staff. There have been approaches on using AI to provide immediate feedback to citizens on their contributions which could be further explored [20].

5.3 First Prototype

We want to describe how a prototype can be designed from the requirements and considerations discussed above. Based on the outcomes of the workshops, a focus group, an online survey, and previous research in the field, we compiled a list of requirements. We then started a prioritization process with all project partners based on the MoSCoW method [57]. These project partners include employees of the city administration of Wuppertal, the company viadukt GmbH, as well as scientific partners with a background in psychology, law, and ethics. Each requirement was rated according to the categories Must, Should, Could, and Won't. Requirements that project partners did not agree on were discussed in an internal workshop. This process led to a list of requirements for the first iteration of development.

This first prototype uses an abstract 3D model that offers rotation, zoom, and the possibility of viewing the area from the top (see the right screenshot in Figure 3). While the location for the mockup was based on a park that was previously subject to participation, the prototype offers ways to participate in a current remodeling project for a square in Wuppertal. The urban planner responsible for this project has requested that our tool be used for one phase of the project. The focus is primarily on climate adaptation since the square in question has little vegetation and almost only sealed surfaces that lead to high temperatures in direct sunlight.

The different questions relevant to the urban planner and city staff (top-down approach to participation) are distributed over several stations which can be visited by clicking on a location or by clicking on the next button. This design of splitting the experience into different parts was developed together with the city administration and software development company. Inspiration was drawn from other e-participation tools, such as the location hunting game by Tabi and Ikeda [109] which combines physical and digital forms of interaction with public spaces to invite users to explore and become familiar with locations, or the mobile walking game by Takahashi et al. [110], which is specifically intended to encourage

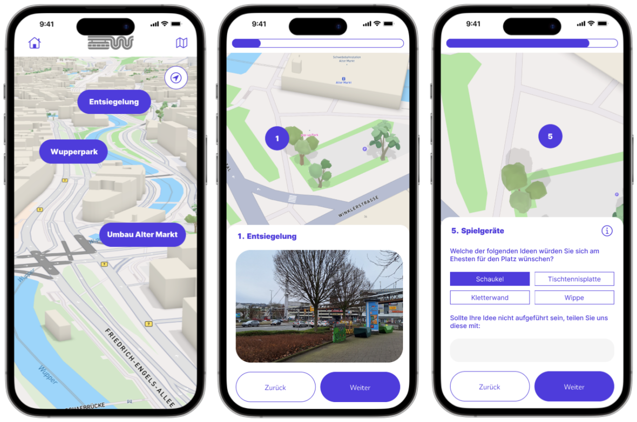

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Figure 3: Three screenshots from the first prototype. The left screenshot shows the "discovery mode" which allows users to scroll through the city map. The center screenshot shows a photo of the square for an activity on unsealing surfaces. The right screenshot shows an activity that allows users to choose from pre-defined answers, or to type in their own answer to a question. The prototype is developed by the company viadukt GmbH.

older adults to leave their home and explore their city. Another source of inspiration was the "commented walks" method by Andersen and Balbontín [5] as a way to experience locations, with a specific focus on sensory and affective information. The aforementioned studies all include physically being at a location, but our goal is to enable users to explore and get familiar with locations both on-site and at home by using an abstract 3D visualization with photos (as opposed to Tabi and Ikeda [109] and Takahashi et al. [110] who use photos and a 2D map to show routes).

Each participation process contains an overview of the most important information, including reasons why the area is being remodeled, and budgetary restrictions (in this case the budget can only be used for climate adaptation measures). In addition, there is a timeline (milestones, important dates, and time left), an FAQ section with information on how to use the tool, and contact details of the responsible person. Users can either first start the participation process to give their own input or directly view input from other users and give it a like. Each station in a participation process contains a photo of the area with a short description of the topic (i.e. unsealing surfaces or adding plants temporarily), and a link to further information. The stations contain different activities, such as rating scales, text input forms, single/multiple choice questions, and photo uploads. If the info button is clicked, users receive a short explanation on how to complete the activity. Users can write in a form to give general feedback on the participation process or the usability of the prototype. In a future development iteration, users will be able to place icons on the 3D model (preference (2) for interaction and communication, see Subsection 4.2). The current phase of the remodeling project does not need feedback on drafts from urban planners yet; however, this is a feature that will be used in the future.

We briefly want to describe some other steps in the development, such as a preliminary evaluation. In a first pre-test with six students, we received some feedback on usability which was compiled into a list and forwarded to the software development company. The design was generally well received and described as being intuitive. Feedback regarding activities was also forwarded to city staff. This feedback included that the surfaces that should be unsealed are not sufficiently marked. We also conducted a first heuristic evaluation [75] with two usability experts based on the "10 Usability Heuristics for User Interface Design" by Nielsen [74]. Findings included that currently there is no button to leave a station and return to the map. The results were summarized in a table with information on the severity of the problem, the affected feature, the relevant heuristic for this problem, details of the problem, and recommendations for improvement, and also forwarded to the company.

We are currently planning three workshops with citizens to show them the current state of development and to test and evaluate this prototype. Two of these workshops will be with the same groups that have previously participated. The third will include citizens that we invite through different channels, such as social media and the city administration's website. After the workshops, the citizens' comments and ideas will inform further development. The city administration also continues to be involved in the development process. We are particularly interested in whether the abstract visualization supported by photos is suitable for the planned activities or whether a more photorealistic visualization is needed at some stations. A new version will be used and further evaluated during the aforementioned remodeling project through interviews and surveys. We are also in contact with associations for persons with disabilities to include them in future tests. There are no AI features implemented, as we are currently researching how to best include them in our prototype.

Beyond concrete features (i.e. the technical system), the broader societal context in which e-participation is embedded needs to be considered. In the civic voluntarism model, Verba [123] argues that a lack of resources and being outside of recruitment networks may lead people to not participate politically. Therefore, offering a new e-participation tool needs to be accompanied by other measures to strengthen the participation of underrepresented groups. The public needs to be made aware of the tool's existence through advertisements in newspapers, at bus stops, and other means (this was also mentioned by our participants). Other ways of raising awareness include contacting well-connected citizens who can inform people in their network. For citizens with less digital literacy, consultation hours in which they can bring their own devices and receive support to participate online should be offered.

While we as researchers can improve and support e-participation processes by including citizens and city staff in the development of tools, the legal basis of citizen participation in Germany remains unchanged, meaning that it remains consultative and that there is no guarantee of seeing individual contributions become reality (see Section 2). In addition, our tool will be used for top-down participation.

At the same time, our findings show that the perception of whether input from citizens is taken into account in decisions is an important motivational factor for participation. The hope is that by offering a well-designed, usable tool, both citizens and city staff will be more inclined to enter into a dialogue and consider each other's opinions. This improved dialogue includes establishing an understanding of what each side needs. Urban planners should be involved in the design process to understand what content a citizen's contribution should have and how it should be structured. In turn, they should be informed about how citizens want to be involved in decision-making processes and what kind of feedback they need to feel that their contributions are meaningful.

All these aspects illustrate that the social system (including organizational factors and people's skills) plays an important role in e-participation. The specific circumstances in our context also make it difficult to directly compare our work to previous research discussed in Section 2. Since one of our goals is to enable participation both on-site and at home, we require a different approach than Takahashi et al. [110] and Tabi and Ikeda [109] who encourage walking/biking to a location, and Kwon et al. [59] whose focus is on face-to-face group design discussions. Our tool is targeted at all citizens and not a special group, such as young people who are the focus of Saßmannshausen et al. [97]. We believe that considering circumstances and goals, rather than offering a techno-centric, generic solution for an e-participation tool, is important.

6 Limitations

The mockup used during the workshop with citizens did not allow for rotation or tilting of the 2D map and the 3D models which might have impacted the participants' experience.

Due to the qualitative nature of the method, not all demographics could be reached. Our participants were recruited through convenience sampling. We are aware of these issues and conducted further studies using different research methods (focus groups, surveys, and experiments) to reach more citizens. Nonetheless, we believe that our findings highlight the differing needs both between and within stakeholder groups, as well as societal and organizational factors that impact e-participation. Although our focus was on Wuppertal, a mid-sized German city, we believe that the description of our research approach and our recommendations can inspire other studies or development processes of tools in other cities, especially those that do not have an e-participation tool yet.

However, recommendations and considerations do not offer a solution to the problem of differing stakeholder needs. We suggest that there is not one solution that fits every municipality, especially smaller ones that have less dedicated staff or financial resources. They are meant as guidance when considering both social and technical factors in the e-participation ecosystem.

7 Summary and Conclusions

This paper discussed ideas for an e-participation tool using a 3D city model derived from co-design workshops with citizens, experts, and city staff. We aimed to explore participants' wishes and needs regarding 1) information, and 2) interaction and communication. We found that while there were some features all groups agreed on, there were some disagreements as well. An example of agreement between stakeholder groups was the need for better ways to categorize and present information. The city staff agrees with the citizens that information and feedback are important for a transparent participation process, and stated their wish for the possibility of better ways for categorizing information and offering it in visual forms. A salient example of a disagreement was the registration process, which citizens found to be overly complicated, whereas experts and city staff agreed that a certain amount of information is necessary to avoid abuse and misconduct.

We found that citizens have different needs when it comes to *how* they want to participate. Some only want to comment and vote, others want to place icons, and still others prefer to sketch to visualize their ideas. These citizens' wishes for fewer restrictions when contributing ideas are also tied to a higher moderation load for city staff. This means when contemplating which features should be implemented, the resource constraints of the municipality need to be considered. If possible, we recommend that different types of participation should be offered to allow citizens to participate in the way most suitable to them.

From the mock-ups presented to participants, the most popular level of realism for a visualization of the city was an abstract block model. While some participants argued that the abstraction could negatively impact recognition of the area, there was a consensus that citizens should have some knowledge about the area which is the focus of participation. This knowledge can be supported through photos and videos of the area, or an additional layer showing the terrain.

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8 Appendix

In the following, the sub-codes for each code are listed:

Information

- Personalization and Notification
- Providing Planning Drafts Online
- Feedback
- General Conditions of Participation
- Language
- Pictures and Photos
- Videos

- Text
- Localization
- Marking of Areas
- Available Data
- Loading Time
- Visualization of Consequences of Drafts
- Similarity to Existing Applications
- Visualization of Intermediate Results
- Clustering and Filtering
- Providing different Variants
- Zoom, Rotation, Perspective
- Geographical Information and Terrain
- Realism and Abstraction

Communication and Interaction

- Placing 3D Objects
- Querying Emotional Associations
- Planning Tool
- Sharing Contributions
- Registration
- Text Comments
- Voice Messages
- Freehand Sketches
- Placing Icons
- Moderating Posts and Hate Speech
- Group Chats and Proximity Chats
- Private Messages
- Social Media
- Live-Chat
- Blog
- Consultation Hours
- Voting for Best Ideas
- Giving a Like/Thumbs Up
- Pros and Cons List